

Homework assignment for 02YELMA

Chapter: Mathematical tools

1. Compute the gradient of the scalar field $f(x, y, z) = xyz$. After calculating the gradient, describe its geometric interpretation at the point $(1, 1, 1)$ and explain how it relates to the rate of change of f in various directions from this point.
2. Calculate the divergence of the vector field $\vec{F}(x, y, z) = (x^2, -y^2, z^2)$. Interpret the physical meaning of your result. Discuss how the divergence at the point $(1, -1, 1)$ reflects the behavior of the vector field in terms of sources, sinks, or neutrality.
3. Determine the curl of the vector field $\vec{F}(x, y, z) = (y, z, x)$. Explain the significance of your result in terms of the field's rotation. Illustrate what the curl at the point $(0, 0, 1)$ suggests about the rotational behavior of the vector field around this point.
4. Prove the following vector identities:
 - $\vec{\nabla} \times (\vec{\nabla} T) = 0$
 - $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{T}) = 0$
5. Are $(\vec{A} \times \vec{B}) \times \vec{C}$ and $\vec{A} \times (\vec{B} \times \vec{C})$ equal? Justify your answer.
6. Evaluate the line integral of the vector field $\vec{F} = x\hat{x} + y\hat{y}$ along the closed path defined by the points $A(0, 1) \rightarrow B(1, 1) \rightarrow C(0, 0) \rightarrow A(0, 1)$.
7. Evaluate the surface integral of the vector field $\vec{F} = x\hat{x} + y\hat{y} + z\hat{z}$ over a sphere with radius 1 centered at $(0, 0, 0)$. If $\vec{F}$ describes the flux of a quantity, describe what information does the integral give us? (Hint: Consider spherical coordinates)
8. A cylinder is positioned on the xy plane and centered on the z -axis, with a height of 3 units and a radius of 1 unit. The mass density of the cylinder is determined by the function $\rho(x, y, z) = x + y + z$. Calculate the total mass of the cylinder using the given mass density function. (Hint: Consider cylindrical coordinates)
9. Use the Divergence Theorem to evaluate the flux of the vector field $\vec{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ outward through the surface of a cube with any vertex at the origin and with side length 1.
10. Use Stokes' Theorem to find the circulation of the vector field $\vec{F} = -y\hat{i} + x\hat{j} + z\hat{k}$ around the circle $x^2 + y^2 = 1$ in the plane $z = 1$.
11. Consider the radial vector field of the form $\vec{V} = kr^{-p}\hat{r}$ in three-dimensional space where k is a constant. Which specific value of p ensures that the total flux through any enclosing concentric sphere remains constant, and that the space itself contains absolutely no sources or sinks away from the origin? Provide a detailed explanation of your reasoning, including the mathematical derivation that leads to your conclusion.

12. Calculate the gradient of a scalar field $f = 1/R$, where R is the magnitude of $\vec{R} = \vec{r} - \vec{r}'$, with respect to primed coordinates. Follow the steps below:
- (i) Express $1/R$ in terms of the Cartesian coordinates.
 - (ii) Compute the gradient $\vec{\nabla}'(1/R)$ by finding the partial derivatives of $1/R$ with respect to x' , y' , and z' . Note: $\vec{\nabla}'$ denotes the gradient with respect to the primed coordinates.
 - (iii) Simplify the result and express the gradient in terms of $\hat{R}$ and R .